Hans William Alexander Hanley 汉斯·汉隶

Natural Language Processing • AI Safety • Multilingual Representation Learning

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I research scalable methods for understanding, evaluating, and improving large language models, with expertise in multilingual representation learning, semantic modeling, and LLM-based evaluation. My research spans computational social science and AI safety, focusing on uncovering patterns in large-scale information ecosystems, measuring model behavior, and improving the reliability of AI systems.

Education

Stanford University **PhD, Computer Science, 2020-2025 | MS, Computer Science, 2020-2024**
GPA: 4.02; Advisor: Zakir Durumeric; Meta Research PhD Fellow; NSF GRFP; Stanford EDGE Doctoral Fellow

University of Oxford **MSc, Statistical Science, 2019-2020 | MSc, Computer Science, 2018-2019**
Distinction (Highest Honors); Daniel M. Sachs Class of 1960 Graduating Scholar at Worcester College

Princeton University **BSE, Electrical Engineering, 2014-2018**
Highest Honors, GPA: 3.982; Information Security & Privacy Concentration; Phi Beta Kappa; Tau Beta Pi; Sigma Xi

Professional Experience

Microsoft AI, Member of Technical Staff **April 2026–Present**
- Developed rubric-based LLM-as-a-judge evaluators for Microsoft Copilot conversations, assessing task completion, intent satisfaction, reasoning complexity, response quality, and safety failures to improve models, system prompts, and products.
- Built LLM classifiers and measurement frameworks linking user feedback and behavior to offline evaluations, identifying representativeness and failure-mode coverage gaps; presented evaluation methodology and strategy to Microsoft AI's CEO.

Meta, Research Scientist, CAS / Central Applied Sciences **March 2025–April 2026**
- Researched supervised fine-tuning (SFT) and persona-conditioned prompting for LLM digital twins modeling individual behavior; used verbalized sampling to increase response diversity and reduce behavioral homogenization.
- Deployed large-scale multilingual and multimodal LLM pipelines for content understanding, creator classification, and political content and integrity-risk evaluation across Instagram and Threads to inform recommender systems.

Google Research, Health AI Student Researcher **Summer 2024**
- Designed LLM agents for structured querying and high-fidelity reasoning over large clinical datasets, with verification layers for failures under distribution shifts.
- Built evaluations of accuracy, safety, and adversarial robustness for LLM-mediated medical data interactions.

Research Highlights & Selected Publications

Tracking the Takes and Trajectories of News Narratives **USENIX Security 2025**
- Built a large-scale narrative-tracking system combining specially fine-tuned (e.g., LoRA) encoder-LLM semantic similarity models and zero-shot stance detection to track 146K narratives across 4K+ news sites and understand narrative pathways (e.g., anti-vaccine and anti-Ukraine influence campaigns).

Across the Firewall: Foreign Media's Influence on Chinese Social Media **PNAS 2025**
- Quantified foreign-media penetration into Chinese platforms using cross-lingual alignment pipelines to detect narrative borrowing on the Russo–Ukrainian War.

Hierarchical Level-Wise News Article Clustering via Multilingual Matryoshka Embeddings **ACL 2025**
- Proposed multilingual Matryoshka embeddings enabling controllable semantic granularity (coarse-to-fine); achieved SOTA on SemEval 2022 Task 8 and built scalable hierarchical clustering pipelines surfacing stories, narratives, and macro-themes.

TATA: Stance Detection via Topic-Agnostic and Topic-Aware Embeddings **EMNLP 2023**
- Developed contrastive embedding models for topic-agnostic and topic-aware stance detection; achieved SOTA zero-shot performance (0.771 F1 on VAST) and improved cross-topic generalization in multi-domain news corpora.

Skills

Programming	Python, Java, C, C++, C#, Go, SQL, R, MATLAB; Git; PyTorch, TensorFlow, Hugging Face
ML/LLM	Transformers; LLM fine-tuning; multilingual modeling; interpretability; feature attribution; contrastive learning; LLM-based evaluation; safety & robustness analysis
Languages	English (native); Mandarin Chinese (HSK Level 5)

Selected Awards & Honors

- CSCW 2025 Best Paper Honorable Mention (Top 4%).
- Rising Stars in Data Science (Stanford/UCSD/UChicago); Meta Research PhD Fellow; NSF Graduate Research Fellowship (GRFP); Stanford EDGE Doctoral Fellow; University of Oxford Daniel M. Sachs Class of 1960 Graduating Scholar.
- Charles Ira Young Memorial Tablet & Medal; SEAS Hayes–Palmer Prize; George B. Wood Legacy Prize.